

For investment strategy 1, the present value is GBP11.91, as follows:

$$PV = -100 + \frac{45}{1.10} + \frac{45}{(1.10)^2} + \frac{45}{(1.10)^3} = 11.91.$$

For investment strategy 2, the present value is also GBP11.91, as follows:

$$PV = -100 + \frac{60}{1.10} + \frac{40}{(1.10)^2} + \frac{32.35}{(1.10)^3} = 11.91.$$

Both investment strategies are valuable in that the present value of the sums of their cash flows are positive. Since the present values are identical, subtracting the present value of strategy 1 from the present value of strategy 2 is zero (i.e., $11.91 - 11.91$).

Either strategy is recommended as both are valuable and have equal present values.

In the following sections, we apply the cash flow additivity principle to three different economic situations and further illustrate the principle of no arbitrage by comparing two economically equivalent strategies in each situation.

Implied Forward Rates Using Cash Flow Additivity

Consider two risk-free discount bonds with different maturities as follows:

One-year bond: $r_1 = 2.50\%$

Two-year bond: $r_2 = 3.50\%$

A risk neutral investor seeking to earn a return on GBP100 over a two-year investment horizon has two possible strategies:

Investment strategy 1: Invest today for two years at a known annualized yield of 3.50 percent. Using Equation 5, we may solve for the future value in two years (FV_2) as follows:

$$\text{GBP}100 = FV_2 / (1+r_2)^2;$$

$$FV_2 = \text{GBP}107.12.$$

Investment strategy 2: Invest today for one year at a known yield of 2.50 percent and reinvest in one year's time for one year at a rate of $F_{1,1}$ (the one-year forward rate starting in one year).

These two strategies are summarized in Exhibit 10: